DM³ Research Programme: Document Structure

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OVERVIEW DOCUMENT

Title

A Generative Operator Framework for Open Problems in Mathematics: The dm³ Research Programme

Abstract

We describe a research programme based on the dm³ operator cycle (Compression–Curvature–Fold–Unfolding–Entropic boundary), which provides a common structural language for analysing three open problems: the Collatz conjecture, the Birch and Swinnerton-Dyer conjecture, and the Navier–Stokes regularity problem.

For each problem, the programme (i) identifies a structural correspondence between the known mathematics and the dm³ operator assignments, (ii) locates the precise formal gap between the structural argument and a complete proof, and (iii) proposes a specific formalisation target in Lean 4 (AXLE repository, github.com/TOTOGT/AXLE).

No problem is claimed to be solved. The contribution of this programme is the identification of a common proof architecture and the precise statement of what remains to be done in each case.

Three companion papers develop each case in detail:

- Paper I: Collatz (discrete arithmetic instance)
- Paper II: BSD (algebraic–analytic instance)
- Paper III: Navier–Stokes (continuous fluid instance)

What is established across all three: The dm³ operator assignments are not analogies. In each case, known mathematical objects (Euler factors, functional equations, viscous dissipation) correspond to specific operators. The structural correspondence is precise and independently checkable.

What is not established: The formal gap in each case — the step from structural correspondence to a complete proof — remains open. The programme identifies this gap exactly.

MSC 2020: 37C70, 11B37, 35Q30, 11G40
Keywords: dm^3 framework, Collatz conjecture, Birch–Swinnerton-Dyer, Navier–Stokes, generative operators, formal verification, Lean 4

PAPER I: COLLATZ

Title

The Collatz Conjecture as a Visibility Problem: A dm^3 Framework Perspective

Abstract

The Collatz conjecture — that every positive integer eventually reaches 1 under the iteration T(n) = n/2 (even) or 3n+1 (odd) — remains unproved. This paper presents a structural analysis of the conjecture within the dm^3 operator framework.

What is established:

(1) *Operator correspondence.* The Collatz map admits a precise dm^3 decomposition: parity halving is the compression operator (C); the 3n+1 step is curvature (K); trajectory turning points are fold events (F); descent into the basin of attraction is unfolding (U); the 1–2–4 cycle is the entropic boundary (E). These assignments are not analogical — each is a specific arithmetic operation.

(2) *Mean contraction.* The geometric mean multiplicative factor per compression–curvature pair is 3/4 < 1. This is a proved fact, not a conjecture. Tao (2019) establishes that almost all orbits attain almost bounded values, formalising the dissipation argument for logarithmic density.

(3) *Uniqueness of the attractor.* The 1–2–4 cycle is the unique cycle compatible with the dm^3 closure condition under the 2–3 interaction. This is argued structurally; it is not an independent proof of uniqueness.

(4) *Precise gap identification.* Three structural proof strategies are presented (dissipation, curvature bounding, fold–stabilisation). Each reduces the conjecture to a specific open sub-problem: the gap between "almost all" and "all" in the dissipation argument; the absence of a worst-case descent bound; and the absence of a formal categorical definition of discrete dm^3 membership.

What is not established: The Collatz conjecture is not proved. Discrete dm^3 membership of the Collatz map is argued but not formally verified. Convergence does not follow from the structural argument alone.

Proposed formalisation target (AXLE Target 5): Lean 4 verification of discrete dm^3 membership for the Collatz map, followed by derivation of convergence from the closure theorems of the extended category.

MSC 2020: 37C70, 11B37
Keywords: Collatz conjecture, dm^3 framework, generative operators, mean contraction, formal verification, Lean 4

PAPER II: BSD

Title

The Birch and Swinnerton-Dyer Conjecture as Entropic Closure: A dm^3 Structural Reinterpretation

Abstract

The Birch and Swinnerton-Dyer conjecture — that the rank of an elliptic curve $E/\mathbb{Q}$ equals the order of vanishing of $L(E,s)$ at $s=1$ — is reinterpreted within the dm^3 operator framework.

What is established:

(1) *Operator correspondence.* The known mathematical objects of BSD admit precise dm^3 assignments: the local Euler product factors are compression (C); the conductor is curvature (K); the functional equation is the fold operator (F); the Modularity Theorem is unfolding (U); the BSD condition itself (analytic rank equals algebraic rank) is the entropic boundary (E). Each assignment corresponds to a specific mathematical object, not a metaphor.

(2) *Known results in operator language.* Kolyvagin's theorem (rank $\leqslant 1$ cases), the analytic continuation via Modularity, and the functional equation are described within this language. No new mathematical content is added to these results; the contribution is the structural organisation.

(3) *Resonant ladder structure.* The BSD heuristic — that partial products $\prod_{p\leqslant X} \#E(\mathbb{F}_p)/p$ grow like $C_E\cdot(\log X)^r$ — is identified as structurally parallel to the Collatz resonant ladder $3^k\cdot n/2^m$. In both cases, local compression and global curvature interact to produce a ladder whose exponent records attractor depth.

(4) *Precise gap identification.* BSD is equivalent to the claim that every elliptic curve is a complete dm^3 system. The gap: no proof that E is always reached for arbitrary rank; the categorical extension to higher-rank curves is not formalised.

What is not established: BSD is not proved. The dm^3 reinterpretation does not add analytic content beyond known results. The theorem on p.XX is stated in structural form only and is not

an independent mathematical result.

Proposed formalisation target (AXLE Target 6): Lean 4 formalisation of the operator assignments for rank-0 and rank-1 curves (where BSD is known), as a test case for the general programme.

MSC 2020: 11G40, 14H52, 11G05
Keywords: Birch–Swinnerton-Dyer conjecture, elliptic curves, L-functions, dm^3 framework, generative operators, formal verification

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PAPER III: NAVIER–STOKES

Title

Navier–Stokes Regularity as Entropic Closure: A dm^3 Structural Reinterpretation

Abstract

The Navier–Stokes regularity problem — whether smooth initial data on $\mathbb{R}^3$ always produces a globally smooth solution — is reinterpreted within the dm^3 operator framework as the continuous counterpart of the Collatz case.

What is established:

(1) *Operator correspondence.* The terms of the Navier–Stokes equations admit dm^3 assignments: viscous dissipation is compression (C); the nonlinear advection term $(u\cdot\nabla)u$ is curvature (K); vortex stretching is the fold operator (F); pressure projection is unfolding (U); the energy equality / enstrophy bound is the entropic boundary (E).

(2) *Energy equality as closure condition.* The energy equality $\|u(t)\|^2 + 2\nu\int_0^t\|\nabla u\|^2\,ds = \|u_0\|^2$ is identified with the dm^3 closure condition: dissipation (C) exactly accounts for the energy introduced by curvature (K). This is a known result (Leray 1934 for weak solutions); the dm^3 language organises it.

(3) *Structural incompatibility of blow-up.* The argument that blow-up is structurally incompatible with dm^3 closure is presented: if the entropic boundary E is reached, the energy cascade is bounded and no singularity forms. This is argued structurally; it is not a proof.

(4) *Precise gap identification.* Three gaps parallel the Collatz gaps: (a) no proof that E is always reached for arbitrary initial data; (b) no proof that the 3-6-9 vortex structure is the unique attractor in the continuous setting; (c) the quantitative enstrophy bound that would turn the structural argument into a regularity theorem is missing.

What is not established: NS regularity is not proved. The structural argument does not constitute a blow-up prevention proof. No new analytic estimates are derived.

Proposed formalisation target (AXLE Target 7): Lean 4 formalisation of the energy equality in dm^3 language, as a foundation for the regularity argument.

MSC 2020: 35Q30, 76D05, 35B65
Keywords: Navier–Stokes equations, regularity, energy equality,
dm^3 framework, generative operators, formal verification

WHAT TO DO WITH THE MONSTER PAPER CONTENT

The current Monster Paper contains significant non-mathematical content (historical lineages, capoeira, tooth regeneration, mitochondrial DNA, institutional theory) that does not belong in the three mathematics papers. Two options:

Option A: Retain the Monster Paper as a separate philosophical/ programmatic document, clearly labelled as such, not as a mathematics paper. Retitle it: "A Generative Architecture for Planetary Systems: Philosophical and Programmatic Foundations of the dm^3 Framework."

Option B: Archive the Monster Paper as a working document and let the three split papers plus the overview document serve as the public-facing work.

The historical and cultural material is not a weakness if it is in the right container. It becomes a weakness when it shares a document with theorems about BSD and Navier–Stokes, because it signals to mathematical readers that the claims are not being made in good faith.